

# Biotech Catalyst Honesty Backtest

R / f(M) / Finding analysis — predicting clinical-catalyst HIT vs MISS from pre-cutoff marketing, verified against the as-of-cutoff authoritative record. Thesis: EXCLUSION of programs whose marketing diverges from the registry / scientific record, not biology forecasting.

Signal OS · public\_biotech vertical · generated 2026-05-31

## Discrimination scorecard

| Trial | Cohort      | Blind call     | gap   | Predicted | Actual | OK |
|-------|-------------|----------------|-------|-----------|--------|----|
| SAVA  | in-sample   | MISS_CANDIDATE | 0.630 | MISS      | MISS   | Y  |
| MDGL  | in-sample   | MISS_CANDIDATE | 0.489 | MISS      | HIT    | N  |
| CRTX  | held-out #1 | MISS_CANDIDATE | 0.278 | MISS      | MISS   | Y  |
| KRTX  | held-out #1 | CLEAN          | 0.103 | HIT       | HIT    | Y  |
| CYTK  | held-out #2 | MISS_CANDIDATE | 0.292 | MISS      | HIT    | N  |
| AVIR  | held-out #2 | MISS_CANDIDATE | 0.429 | MISS      | MISS   | Y  |

**Binary gate now discriminates on 4 of 6 (SAVA, CRTX, KRTX, AVIR correct; MDGL, CYTK are false positives on HITs).** After the construct-gate fix, KRTX flips to CLEAN and CRTX keeps its earned MISS. Both remaining errors are the SAME mechanism — the *open\_label\_extrapolation* channel grading open-label supporting-color citations as severe — not the endpoint channel that was fixed.

## What the six trials test

**In-sample matched pair (SAVA / MDGL).** Cassava/simufilam (Alzheimer's Phase 3) vs Madrigal/resmetirom (MASH Phase 3). Establishes the plumbing — mechanically enforced blinding, as-of-cutoff M reconstruction, R-vs-M adjudication. SAVA carries a fraud-adjacent overlay (\$40M SEC settlement), so its MISS is a strong-signal case; MDGL is a known false positive driven by the open-label channel.

**Held-out pair #1 (CRTX / KRTX).** Cortexyme/atuzaginstat (GAIN) vs Karuna/KarXT (EMERGENT-2). Both CNS, neither with an integrity overlay. This pair first exposed the gate bug: CRTX was an earned MISS on a real co-primary construct swap, but KRTX (a HIT) was false-excluded on a one-week timeFrame restatement.

**The fix (implemented; evaluator otherwise frozen).** A SEVERE endpoint refute is now admitted to the decisive set only when the registry version history shows a genuine construct/type change of the success-slot primary; timeFrame/timepoint and wording deltas are demoted. This flips KRTX to CLEAN and leaves CRTX a MISS.

**Held-out pair #2 (CYTK / AVIR) — the real test of the fix.** Cytokinetics/aficamten (SEQUOIA-HCM, HIT) vs Atea/AT-527 (MORNINGSKY, MISS), run with the fixed frozen evaluator. The endpoint channel behaved correctly on both (CYTK construct unchanged → endpoint silent; AVIR construct flagged). But CYTK was **still** false-excluded — by 7 *open\_label\_extrapolation* refutes on its FOREST-HCM open-label extension, the identical MDGL failure mode. The held-out HIT cleanly isolates the next channel to fix, and confirms the endpoint fix itself works.

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## Cassava Sciences, Inc. (SAVA) — simufilam

Alzheimer's disease (mild-to-moderate) · phase3\_topline · blind cutoff 2024-11-24 · cohort: in-sample

Blind call: **MISS\_CANDIDATE (EXCLUDE)** · honesty\_gap = **0.630** · decisive refutes = **57** · verdicts: REFUTED 101 / UNVERIFIABLE 277 / AFFIRMED 71

### ***Decisive flags (basis for exclusion)***

**[evidence\_quality\_overreach]** bio.marketing\_exceeds\_source

**R (claim):** “patients with mild Alzheimer s disease who received simufilam treatment continuously for two years (n=47) had no decline in ADAS-Cog scores ( 1.51 SE) as a group”

**Finding:** The claim presents open-label, uncontrolled data (-1.51 ADAS-Cog change) as evidence of efficacy ('no decline') without a control group. This is from NCT04388254, which had open-label periods. The scientific source (2020 paper) reports open-label biomarker results but lacks controlled efficacy validation. Marketing an uncontrolled effect as efficacy evidence constitutes overreach.

**[evidence\_quality\_overreach]** bio.marketing\_exceeds\_source

**R (claim):** “Patients with mild Alzheimer s who received simufilam treatment non-continuously for two years (n=40) declined 1 point on ADAS-Cog ( 1.65 SE) as a group.”

**Finding:** The claim presents open-label, uncontrolled data (-1.65 ADAS-Cog change, 'declined 1 point') from NCT04388254 as evidence of treatment effect. This is an open-label study period result without a concurrent control, marketed as if it demonstrates efficacy.

**[evidence\_quality\_overreach]** bio.marketing\_exceeds\_source

**R (claim):** “Mild patients who received simufilam for 24 continuous months (n=47) showed an average change of 0.07 points on ADAS-Cog11 ( 1.51 SE), baseline to month 24, as a group.”

**Finding:** The claim presents open-label data (-0.07 ADAS-Cog11 change, 'showed no decline') from NCT04388254 continuous treatment as evidence of efficacy. This is uncontrolled, single-arm data marketed as demonstrating treatment benefit without a control comparison.

**Unsealed outcome (2024-11-25): MISS (-84%)** — RETHINK-ALZ (NCT04994483) failed BOTH co-primary endpoints (ADAS-Cog12 and ADCS-ADL); no statistically significant cognitive or functional benefit vs placebo. REFOCUS-ALZ and the OLE were subsequently terminated.

**Prediction CORRECT.**

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## Madrigal Pharmaceuticals, Inc. (MDGL) — resmetirom

MASH / NASH with fibrosis · phase3\_topline · blind cutoff 2022-12-18 · cohort: in-sample

Blind call: **MISS\_CANDIDATE (EXCLUDE)** · honesty\_gap = **0.489** · decisive refutes = **7** · verdicts: REFUTED 19 / UNVERIFIABLE 62 / AFFIRMED 9

### *Decisive flags (basis for exclusion)*

**[endpoint\_manipulation]** bio.endpoint\_integrity

**R (claim):** “The dual primary surrogate endpoints on biopsy are NASH resolution with 2-point reduction in NAS (NAFLD Activity Score), and with no worsening of fibrosis OR a 1-point decrease in fibrosis with no wor”

**Finding:** The claim describes 'dual primary surrogate endpoints' as 'NASH resolution with 2-point reduction in NAS...OR a 1-point decrease in fibrosis...' and states 'Achievement of either primary endpoint is considered a successful trial outcome.' The registered primary as of the cutoff (version 8, 2022-11-04) is 'Week 52 Dual Primary Objectives: To determine the effect of 80 or 100 mg MGL-3196 vs matching placebo on liver bi

**[endpoint\_manipulation]** bio.endpoint\_integrity

**R (claim):** “The two primary surrogate endpoints on biopsy will be NASH resolution, with at least a 2-point reduction in NAS (NASH Activity Score), and with no worsening of fibrosis OR a one point decrease in fibr”

**Finding:** The claim describes 'two primary surrogate endpoints on biopsy' as 'NASH resolution...OR a one point decrease in fibrosis...' implying two distinct alternative primaries. The registered primary as of the claim's filing date (2022-05-09) and as of the cutoff is 'Week 52 Dual Primary Objectives: To determine the effect of 80 or 100 mg MGL-3196 vs matching placebo on liver biopsy (NASH CRN score) at Week 52 compared wit

**[endpoint\_manipulation]** bio.endpoint\_integrity

**R (claim):** “The primary surrogate endpoint on biopsy will be NASH resolution, with at least a 2-point reduction in NAS (NASH Activity Score), and with no worsening of fibrosis.”

**Finding:** The claim describes 'the primary surrogate endpoint on biopsy' (singular) as 'NASH resolution, with at least a 2-point reduction in NAS...and with no worsening of fibrosis,' omitting the alternative fibrosis-improvement criterion. However, the registered Week 52 primary for NCT03900429 is 'To determine the effect of 80 or 100 mg MGL-3196 vs matching placebo on liver biopsy (NASH CRN score) at Week 52'—a single compos

**Unsealed outcome (2022-12-19): HIT (+268%)** — MAESTRO-NASH (NCT03900429) met BOTH co-primary endpoints at Week 52 (NASH resolution and  $\geq 1$ -stage fibrosis improvement) at both 80mg and 100mg doses; statistically significant vs placebo. Led to accelerated approval (Rezdiffra) 2024-03-14.

**Prediction WRONG (false positive on the HIT).**

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## Cortexyme, Inc. (CRTX) — atuzaginstat (COR388)

Alzheimer's disease (mild-to-moderate) · phase2\_3\_topline · blind cutoff 2021-10-25 · cohort: held-out #1

Blind call: **MISS\_CANDIDATE (EXCLUDE)** · honesty\_gap = 0.278 · decisive refutes = 2 · verdicts: REFUTED 3 / UNVERIFIABLE 22 / AFFIRMED 11

### ***Decisive flags (basis for exclusion)***

[endpoint\_manipulation] bio.endpoint\_integrity

**R (claim):** “the co-primary endpoints for the GAIN Trial s interim analysis are change from baseline in ADAS-Cog11 and CDR-SB versus placebo”

**Finding:** The claim (filed 2020-05-12) states the co-primary endpoints for the interim analysis are ADAS-Cog11 and CDR-SB. However, the primary\_endpoint\_timeline shows that as of version 11 (2020-02-18, the registry version active when the claim was filed), the registered co-primaries were ADAS-Cog11 and CDR-SB. But by version 19 (2021-02-18, before the catalyst cutoff), CDR-SB was replaced with ADCS-ADL as the co-primary. The

[endpoint\_manipulation] bio.endpoint\_integrity

**R (claim):** “The co-primary endpoints for both analyses will be change from baseline in ADAS-Cog11 and CDR-SB versus placebo”

**Finding:** The claim (filed 2020-03-17) states the co-primary endpoints for both analyses are ADAS-Cog11 and CDR-SB. At filing date, the registry version 11 (2020-02-18) did register ADAS-Cog11 and CDR-SB as co-primaries. However, version 19 (2021-02-18) changed the co-primary from CDR-SB to ADCS-ADL before the catalyst cutoff. This represents a mid-trial construct change (different functional assessment scale) of a registered

**Unsealed outcome (2021-10-26): MISS (-70%)** — GAIN (NCT03823404) failed both co-primary endpoints (ADAS-Cog11 and ADCS-ADL); no statistically significant cognitive or functional benefit vs placebo in the overall population. Company highlighted a pre-specified P. gingivalis DNA-positive subgroup (n=242) showing ~57% slowing on ADAS-Cog11 (p=0.02), but the trial did not meet its primary objective; program

**Prediction CORRECT.**

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## Karuna Therapeutics, Inc. (KRTX) — KarXT (xanomeline-trospium)

Schizophrenia (acute psychosis) · phase3\_topline · blind cutoff 2022-08-07 · cohort: held-out #1

Blind call: **CLEAN (HIT-consistent)** · honesty\_gap = 0.103 · decisive refutes = 0 · verdicts: REFUTED 3 / UNVERIFIABLE 35 / AFFIRMED 20

### ***Decisive flags (basis for exclusion)***

No decisive high-precision refutes.

**Unsealed outcome (2022-08-08): HIT (+70%)** — EMERGENT-2 (NCT04659161) met its primary endpoint: KarXT reduced PANSS total score by -9.6 points vs placebo at Week 5 (p<0.0001), with significant separation on key secondaries. Statistically significant and clinically meaningful; led to FDA approval (Cobenfy) 2024-09-26.

**Prediction CORRECT.**

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## Cytokinetics, Incorporated (CYTK) — aficamten (CK-274)

Obstructive hypertrophic cardiomyopathy (oHCM) · phase3\_topline · blind cutoff 2023-12-26 · cohort: held-out #2

Blind call: **MISS\_CANDIDATE (EXCLUDE)** · honesty\_gap = **0.292** · decisive refutes = **7** · verdicts: REFUTED 11 / UNVERIFIABLE 81 / AFFIRMED 21

### ***Decisive flags (basis for exclusion)***

**[evidence\_quality\_overreach]** bio.open\_label\_extrapolation

**R (claim):** “Long-Term Data from FOREST-HCM, the Open-Label Extension Study of Aficamten, Show Sustained Improvements in Clinical Efficacy Endpoints”

**Finding:** The claim cites 'sustained improvements in clinical efficacy endpoints' from FOREST-HCM, an open-label extension study, and generalizes this as evidence. The catalyst trial NCT05186818 is randomized, double-masked, parallel-group phase 3. Uncontrolled improvement in a progressive/relapsing cardiac disease (oHCM) is subject to placebo effect and regression to the mean; the open-label data cannot establish efficacy for

**[evidence\_quality\_overreach]** bio.open\_label\_extrapolation

**R (claim):** “longer-term data from FOREST-HCM that demonstrate durable reductions in pressure gradients and cardiac biomarkers as well as improved symptoms in patients with obstructive HCM”

**Finding:** The claim presents 'durable reductions in pressure gradients and cardiac biomarkers as well as improved symptoms' from FOREST-HCM open-label extension as demonstration of efficacy. The catalyst trial NCT05186818 is randomized and double-masked. Open-label improvements in oHCM (a relapsing condition with fluctuating gradients) cannot be generalized as efficacy without a placebo-controlled comparison.

**[evidence\_quality\_overreach]** bio.open\_label\_extrapolation

**R (claim):** “demonstrating sustained reductions in left ventricular outflow tract (LVOT) gradients with no treatment interruptions for low left ventricular ejection fraction (LVEF) due to aficamten”

**Finding:** The claim asserts 'sustained reductions in left ventricular outflow tract (LVOT) gradients' from FOREST-HCM open-label extension. The catalyst trial NCT05186818 is randomized, double-masked, parallel-group. Uncontrolled LVOT gradient reductions in oHCM cannot establish efficacy for the controlled catalyst; LVOT gradients fluctuate and are susceptible to placebo effect and regression to the mean.

**Unsealed outcome (2023-12-27): HIT (+9%)** — SEQUOIA-HCM (NCT05186818) met its primary endpoint: aficamten produced a statistically significant, clinically meaningful improvement in peak oxygen uptake (pVO2) vs placebo at Week 24 ( $p < 0.0001$ ), and met all 10 prespecified secondary endpoints. Supported NDA/MAA filings.

**Prediction WRONG (false positive on the HIT).**

## Atea Pharmaceuticals, Inc. (AVIR) — AT-527 (bemnifosbuvir)

COVID-19 (outpatient, mild-to-moderate) · phase3\_topline · blind cutoff 2021-10-18 · cohort: held-out #2

Blind call: **MISS\_CANDIDATE (EXCLUDE)** · honesty\_gap = **0.429** · decisive refutes = **1** · verdicts: REFUTED 3 / UNVERIFIABLE 8 / AFFIRMED 3

### Decisive flags (basis for exclusion)

[endpoint\_manipulation] bio.endpoint\_integrity

**R (claim):** "The primary endpoint of this trial is time to alleviation or improvement of COVID-19 symptoms maintained for 24 hours."

**Finding:** The claim states the primary endpoint is 'time to alleviation or improvement of COVID-19 symptoms maintained for 24 hours' (filed 2021-08-12). The registered primary endpoint as of the version closest to that date (v2, 2021-07-09) is 'Time to Alleviation or Improvement of COVID-19 Symptoms (21.5 hours)' — not 24 hours. The marketed timepoint (24h) contradicts the registered construct specification (21.5h), representi

**Unsealed outcome (2021-10-19): MISS (-19%)** — MORNINGSKY (NCT04889040) failed its primary endpoint: AT-527 did not achieve a statistically significant reduction in time to symptom alleviation in the overall outpatient population. Atea highlighted numerical benefit in a high-risk subgroup, but the trial missed; Roche later returned rights. Phase 3 program in outpatients was discontinued.

Prediction **CORRECT**.

### Why one endpoint flag was right and one was wrong

Both CRTX and KRTX were excluded by the same rule (bio.endpoint\_integrity, graded SEVERE → decisive). The rule is meant to fire only on a genuine construct/type change of the success-determining primary endpoint. On CRTX it did exactly that; on KRTX it fired on noise.

|                         | Cortexyme GAIN (correct)                               | Karuna EMERGENT-2 (false positive)           |
|-------------------------|--------------------------------------------------------|----------------------------------------------|
| Marketed primary        | ADAS-Cog11 + CDR-SB co-primaries                       | PANSS total at Week 6                        |
| Registry (as-of-cutoff) | v11 had CDR-SB; v19 replaced it with ADCS-ADL          | PANSS total at Week 5 in every version       |
| Nature of mismatch      | Construct swap: dementia-staging scale -> daily-living | Same week timeFrame, but construct identical |
| Correct grade           | SEVERE / decisive (real masking + instability signal)  | Minor at most; NOT decisive                  |
| Outcome                 | MISS (-70%) — flag paid off                            | HIT (+70%) — flag was wrong                  |

**Fix (not yet applied).** A decisive endpoint refute should require a genuine construct/type change of the success-slot primary — verified against registry version history — and must treat pure timeFrame/wording deltas (Week-5-vs-6) and near-term staggered-co-primary emphasis as non-decisive. The fix must be validated on a fresh held-out name; re-running KRTX after tuning to it would repeat the in-sample teaching-to-the-test trap.